Lösung: lineare Beschleunigung

a) $a(t) = -g + \alpha \cdot t$

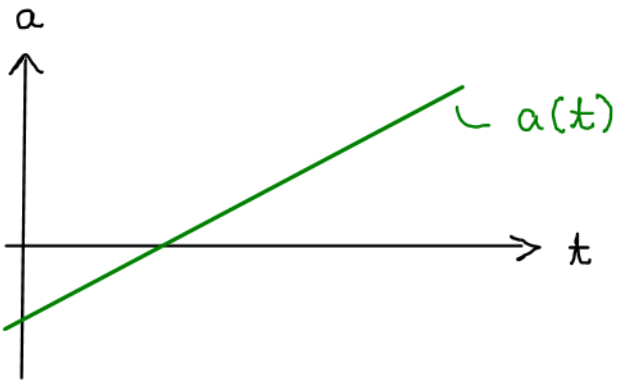
$\frac{m}{s^2} = [a(t)] = [g] = [\alpha][t]$

$\boxed{[g] = m/s^2}$

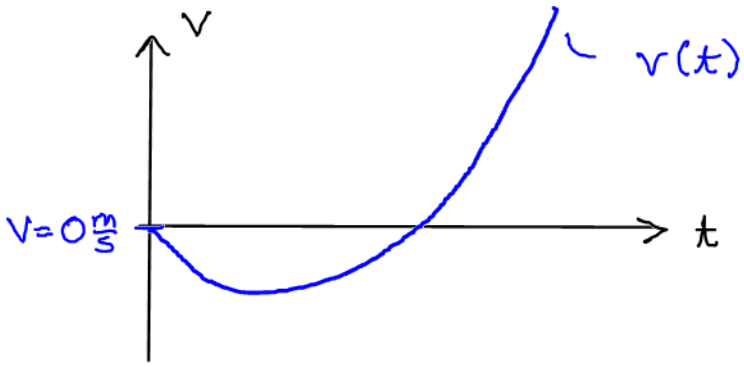
$[\alpha] \cdot s = m/s^2$

$\Rightarrow \boxed{[\alpha] = m/s^3}$

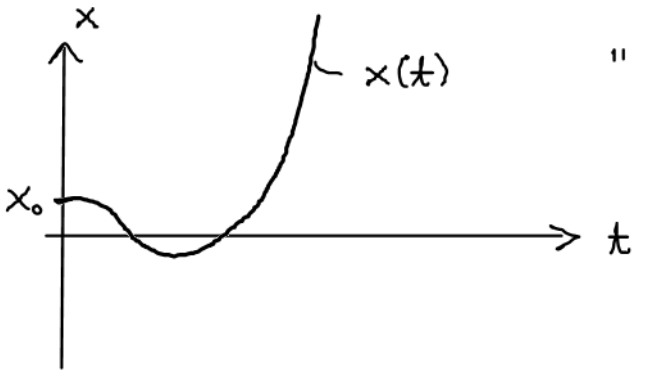
b)



"lineare Funktion"



"Parabel"



"kubische Funktion"

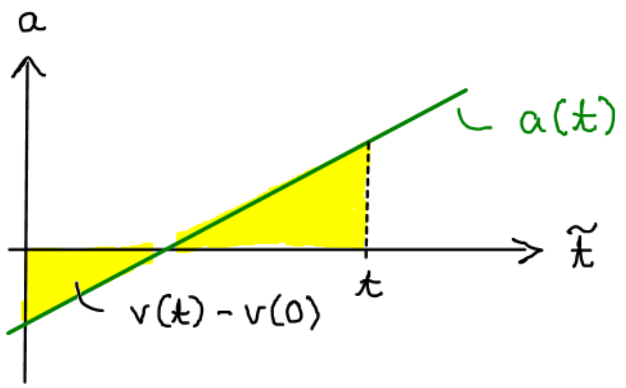
c) Geschw. $v(t)$: Fläche unter dem $a(t)$-Graphen

$$v(t) = \overbrace{v(0)}^{=0\,\frac{m}{s}} + \int_0^t a(\tilde{t})\,d\tilde{t}$$

$$= \int_0^t (-g + \alpha\tilde{t})\,d\tilde{t}$$

$$= \left[-g\tilde{t} + \frac{\alpha}{2}\cdot\tilde{t}^2\right]_0^t$$

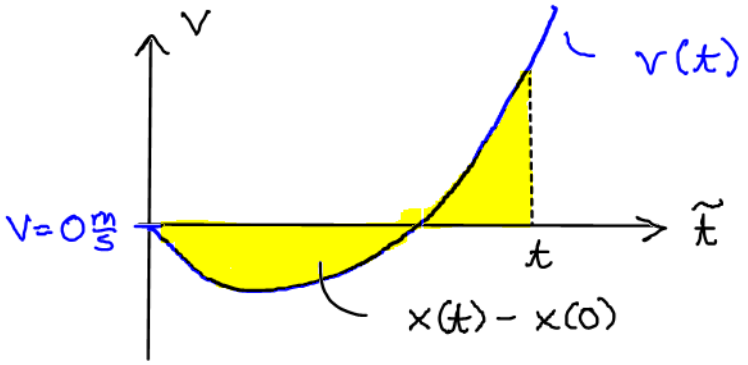
$$= -gt + \frac{\alpha}{2}t^2$$



"gesteigert" beschleunigte Bewegung

gleichmäßig beschleunigte Bewegung

Distanz $x(t)$: Fläche unter dem $v(t)$-Graphen



$$x(t) = \overbrace{x(0)}^{=x_0} + \int_0^t v(\tilde{t})\,d\tilde{t}$$

$$= x_0 + \int_0^t \left(-g\tilde{t} + \frac{\alpha}{2}\tilde{t}^2\right) d\tilde{t}$$

$$= x_0 + \left[-\frac{g}{2}\tilde{t}^2 + \frac{\alpha}{6}\tilde{t}^3\right]_0^t$$

$$= x_0 - \frac{g}{2}t^2 + \frac{\alpha}{6}\tilde{t}^3$$

"gesteigert" beschleunigte Bewegung

gleichmäßig beschleunigte Bewegung

Anfangsposition